Lab Experiment 7: CI/CD using Jenkins, GitHub and Docker Hub

1. Aim

To design and implement a complete CI/CD pipeline using Jenkins, integrating source code from GitHub, and building & pushing Docker images to Docker Hub.

2. Objectives

- Understand CI/CD workflow using Jenkins (GUI-based tool)
- Create a structured GitHub repository with application + Jenkinsfile
- Build Docker images from source code
- Securely store Docker Hub credentials in Jenkins
- Automate build & push process using webhook triggers
- Use same host (Docker) as Jenkins agent

3. Theory

What is Jenkins?
Jenkins is a web-based GUI automation server used to:
* Build applications
* Test code
* Deploy software

It provides:
* Dashboard (browser-based UI)
* Plugin ecosystem (GitHub, Docker, etc.)
* Pipeline as Code using Jenkinsfile

What is CI/CD?
* Continuous Integration (CI): Code is automatically built and tested after each commit
* Continuous Deployment (CD): Built artifacts (Docker images) are automatically delivered/deployed

Workflow Overview
Developer -> GitHub -> Webhook -> Jenkins Build -> Docker Hub

4. Prerequisites

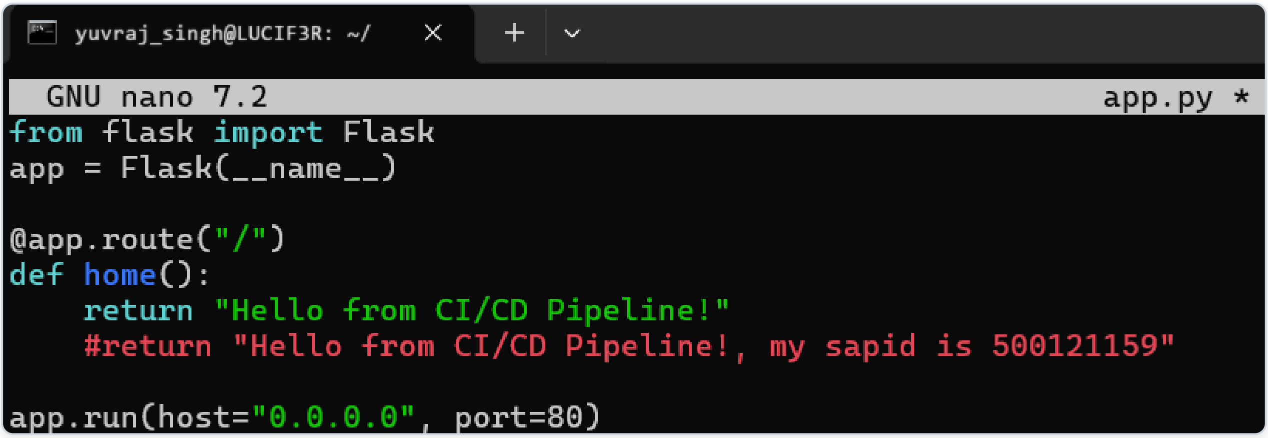
- Docker & Docker Compose installed
- GitHub account
- Docker Hub account
- Basic Linux command knowledge

5. Part A: GitHub Repository Setup (Source Code + Build Definition)

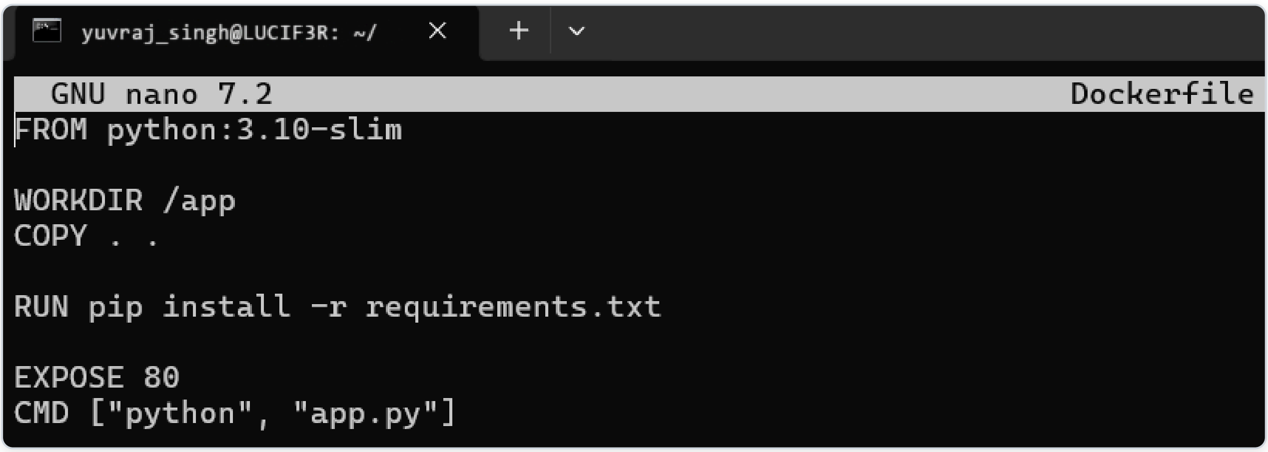
5.1 Create Repository

Create a repository on GitHub: my-app

5.2 Application Code



5.3 Dockerfile (Build Process)



Build Process Explanation

1. Source code pushed to GitHub
2. Jenkins pulls code
3. Dockerfile:

* Creates environment
* Installs dependencies
* Packages app

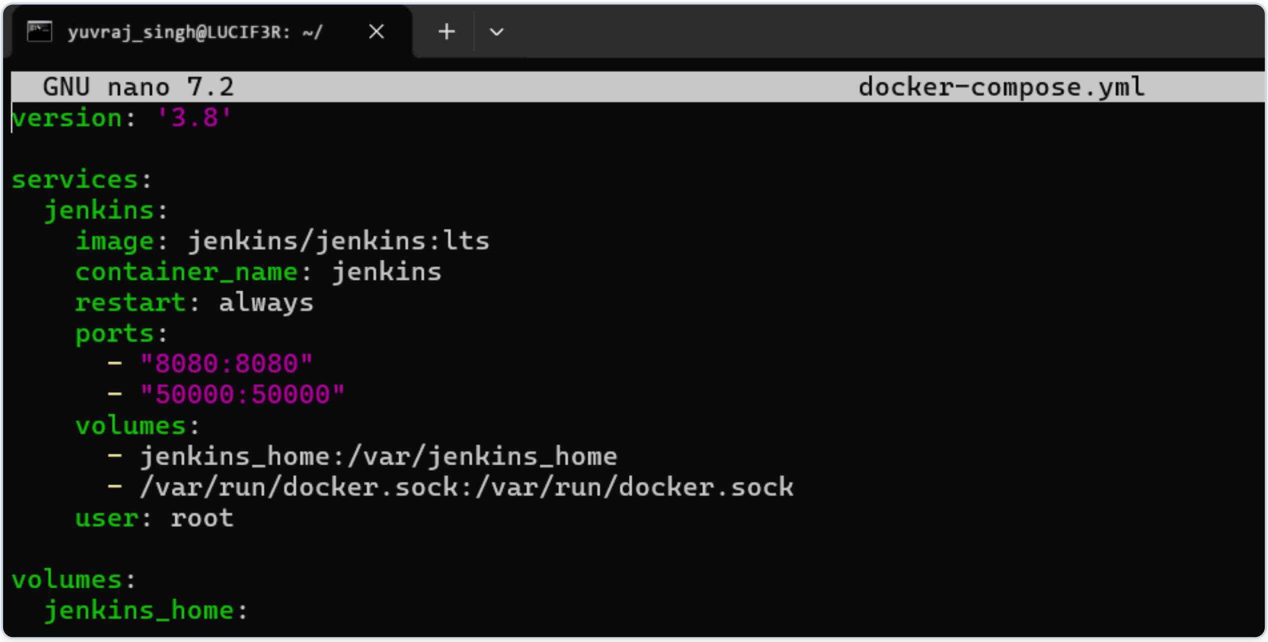
4. Output Docker Image

5.4 Jenkinsfile (Pipeline Definition in GitHub)

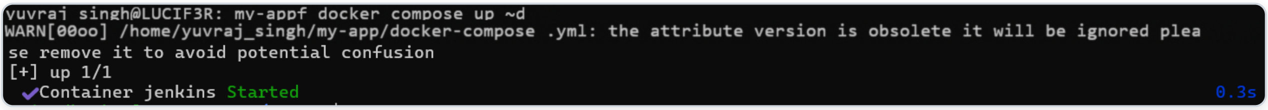
```
yuvraj_singh@LUCIF3R: ~/
  GNU nano 7.2                                              Jenkinsfile
pipeline {
    agent any

    environment {
        IMAGE_NAME = "your-dockerhub-username/myapp"
    }

    stages {

        stage('Clone Source') {
            steps {
                git 'https://github.com/your-username/my-app.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t $IMAGE_NAME:latest .'
            }
        }

        stage('Login to Docker Hub') {
            steps {
                withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
                    sh 'echo $DOCKER_TOKEN | docker login -u your-dockerhub-username --password-stdin'
                }
```

6. Part B: Jenkins Setup using Docker (Persistent Configuration)

6.1 Create Docker Compose File



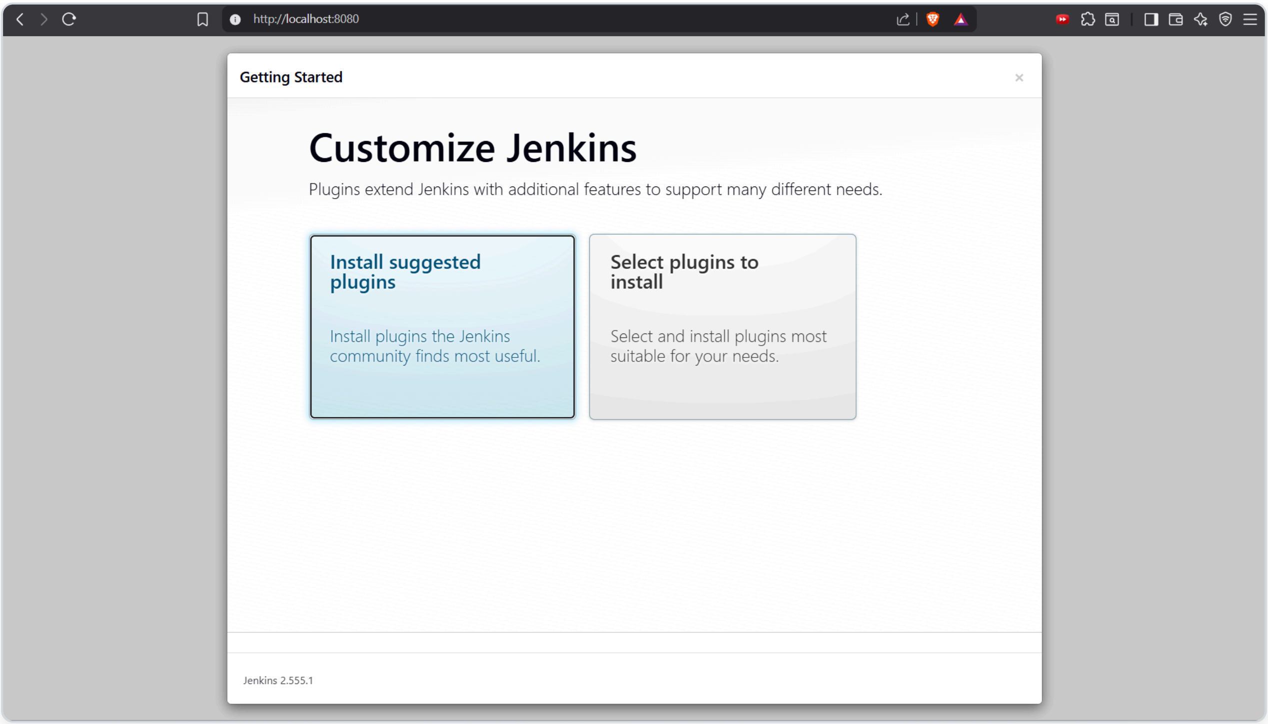
6.2 Start Jenkins



Access: http://localhost:8080

6.3 Unlock Jenkins

docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword



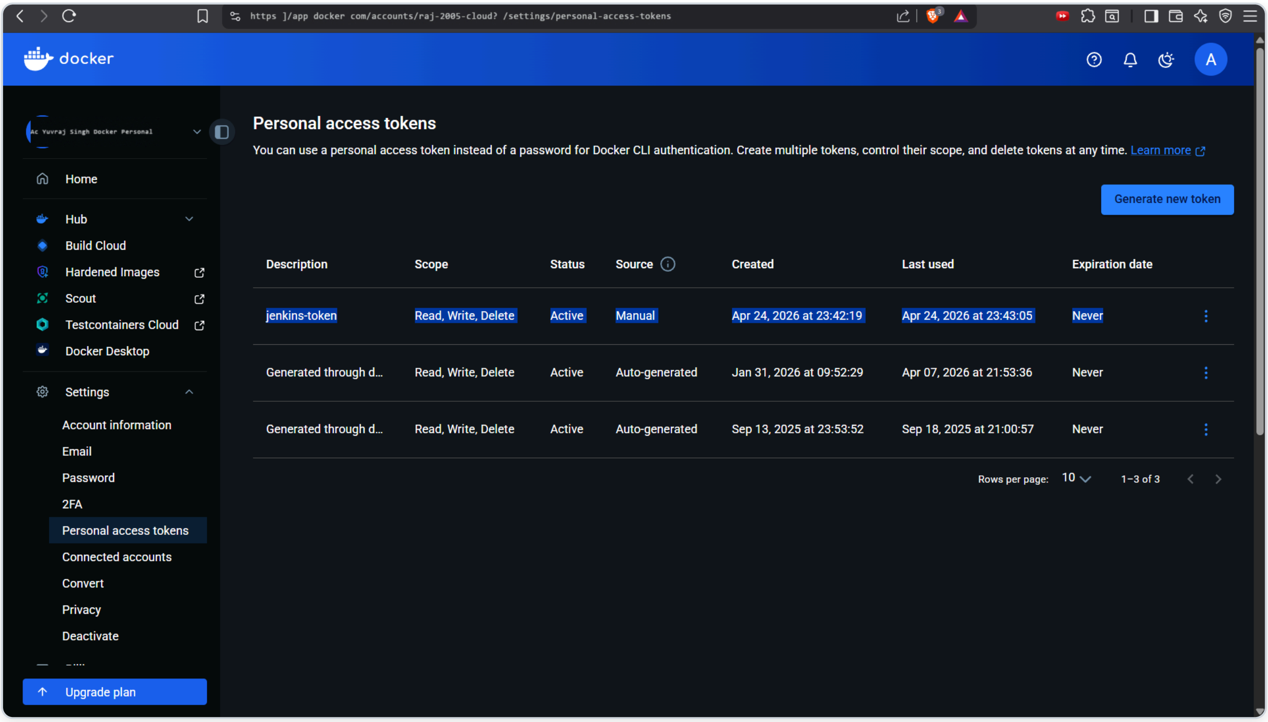
6.4 Initial Setup

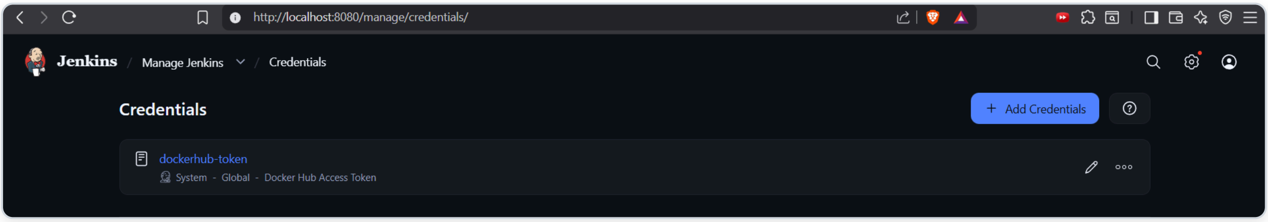
- Install suggested plugins
- Create admin user

7. Part C: Jenkins Configuration

7.1 Add Docker Hub Credentials

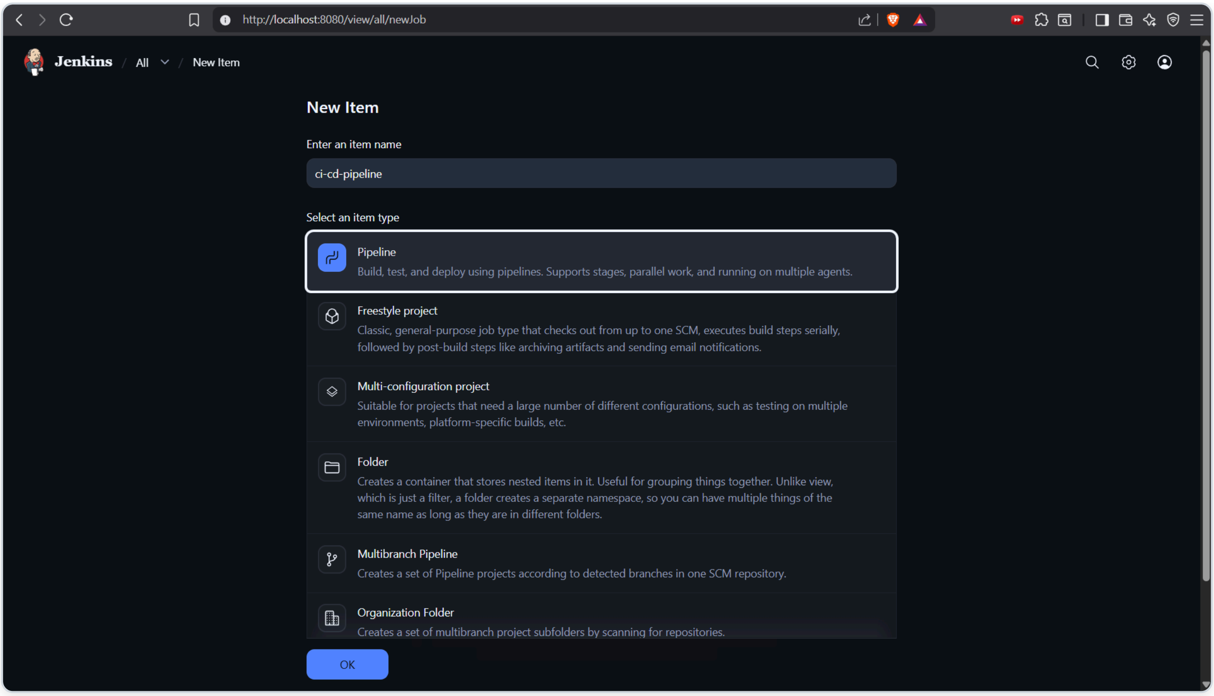
Path: Manage Jenkins -> Credentials -> Add Credentials
* Type: Secret Text
* ID: dockerhub-token
* Value: Docker Hub Access Token





7.2 Create Pipeline Job

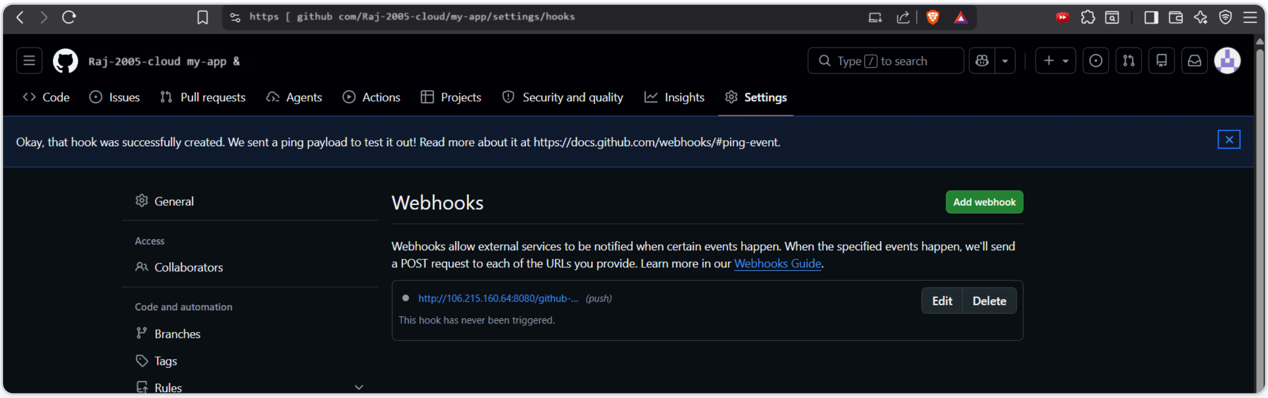
1. New Item -> Pipeline
2. Name: ci-cd-pipeline
3. Configure:
   * Pipeline script from SCM
   * SCM: Git
   * Repo URL: your GitHub repo
   * Script Path: Jenkinsfile



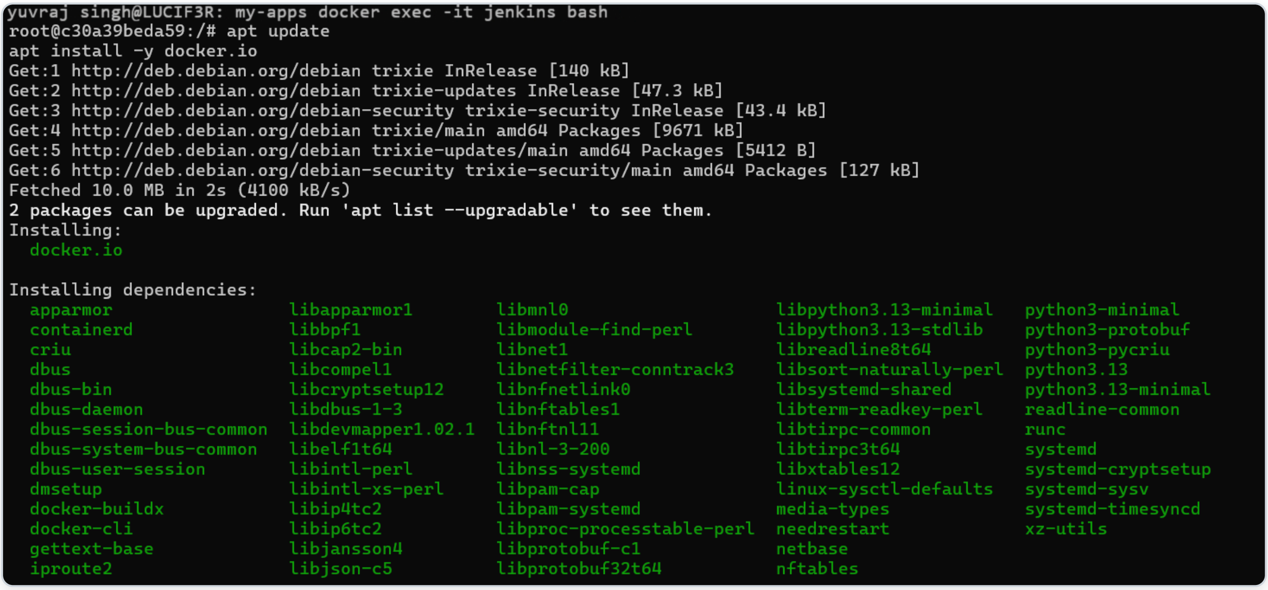
8. Part D: GitHub Webhook Integration

8.1 Configure Webhook

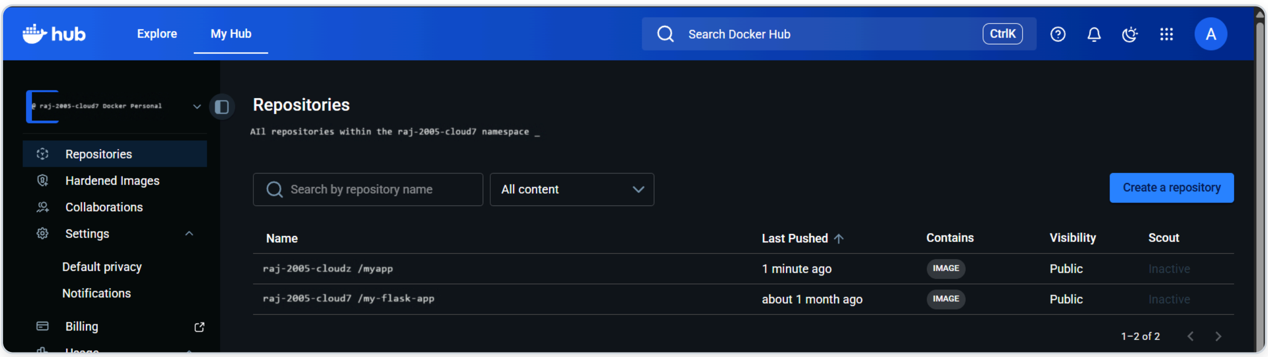
In GitHub: Settings -> Webhooks -> Add Webhook
* Payload URL: http://:8080/github-webhook/
* Events: Push events



DEMO


yuvraj singh@LUCIF3R: my-apps docker exec -it jenkins bash
root@c30a39beda59:/# apt update
apt install -y docker.io
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [127 kB]
Fetched 10.0 MB in 2s (4100 kB/s)
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
docker.io
Installing dependencies:
apparmor libapparmor1 libmnl0 libpython3.13-minimal python3-minimal
containerd libbpf1 libmodule-find-perl libpython3.13-stdlib python3-protobuf
criu libcap2-bin libnet1 libreadline8t64 python3-pycriu
dbus libcompel1 libnetfilter-conntrack3 libsort-naturally-perl python3.13
dbus-bin libcryptsetup12 libnfnetlink0 libsystemd-shared python3.13-minimal
dbus-daemon libdbus-1-3 libnftables1 libterm-readkey-perl readline-common
dbus-session-bus-common libdevmapper1.02.1 libnftnl11 libtirpc-common runc
dbus-system-bus-common libelf1t64 libnl-3-200 libtirpc3t64 systemd
dbus-user-session libintl-perl libnss-systemd libxtables12 systemd-cryptsetup
dmsetup libintl-xs-perl libpam-cap linux-sysctl-defaults systemd-sysv
docker-buildx libip4tc2 libpam-systemd media-types systemd-timesyncd
docker-cli libip6tc2 libproc-processtable-perl needrestart xz-utils
gettext-base libjansson4 libprotobuf-c1 netbase
iproute2 libjson-c5 libprotobuf32t64 nftables


http://localhost:8080/job/ci-cd-pipeline/10/
Jenkins
ci-cd-pipeline
#10
Status
Changes
Console Output
Edit Build Information
Delete build '#10'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#10 (Apr 24, 2026, 7:24:48 PM)
Add description
Keep this build forever
Started by user Yuvraj Singh
Started 40 sec ago
Took 38 sec
This run spent:
13 ms waiting;
38 sec build duration;
38 sec total from scheduled to completion.
refs/remotes/origin/main
No changes.


hub
Explore
My Hub
Search Docker Hub
CtrlK
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Repositories
All repositories within the raj-2005-cloud7 namespace
Search by repository name
All content
Create a repository
Name
Last Pushed
Contains
Visibility
Scout
raj-2005-cloud7 /myapp
1 minute ago
IMAGE
Public
Inactive
raj-2005-cloud7 /my-flask-app
about 1 month ago
IMAGE
Public
Inactive
1–2 of 2

9. Part E: Execution Flow (Stage-wise Explanation)

- Stage 1: Code Push - Developer updates code in GitHub
- Stage 2: Webhook Trigger - GitHub sends event to Jenkins
- Stage 3: Jenkins Pipeline Execution
- Stage: Clone - Pulls latest code from GitHub
- Stage: Build - Docker builds image using Dockerfile
- Stage: Auth - Jenkins logs into Docker Hub using stored token
- Stage: Push - Image pushed to Docker Hub
- Stage 4: Artifact Ready - Docker image available globally

10. Role of Same Host Agent

- Jenkins runs inside Docker
- Docker socket mounted: /var/run/docker.sock
- Effect: Jenkins directly controls host Docker. Builds and pushes images without separate agent.

11. Observations

- Jenkins GUI simplifies CI/CD management
- GitHub acts as source + pipeline definition
- Docker ensures consistent builds
- Webhook enables automation

12. Result

Successfully implemented a complete CI/CD pipeline where:
* Source code and pipeline are maintained in GitHub
* Jenkins automatically detects changes
* Docker image is built on host agent
* Image is securely pushed to Docker Hub

13. Viva Questions

1. What is the role of Jenkinsfile?
2. How does Jenkins integrate with GitHub?
3. Why is Docker used in CI/CD?

4. What is a webhook?
5. Why store Docker Hub token in Jenkins credentials?
6. What is the benefit of using same host as agent?

14. Key Notes

- Jenkins is GUI-based but pipeline is code-driven
- Always use credentials store (never hardcode secrets)
- Webhook makes CI/CD fully automatic
- This setup is ideal for learning and small deployments

Understanding Jenkins Pipeline Syntax (Simplified Explanation)

Jenkins pipelines (written in a Jenkinsfile) follow a clear structure of blocks and steps.

1. Basic Pipeline Structure

   pipeline {
   agent any
   stages {
   stage('Build') {
   steps {
   sh 'echo Hello'
   }
   }
   }
   }

2. Key Terms Explained
   * pipeline {} - Root block. Everything is written inside this.
   * agent any - Defines where the pipeline runs (any available node).
   * stages {} - Groups all phases of pipeline. Logical separation (Clone, Build, Test, Deploy).
   * stage('Name') - A single step/phase in pipeline. Visible in Jenkins GUI as blocks.
   * steps {} - Contains actual commands to execute.

Common Steps
* git 'https://github.com/user/repo.git' - Clones source code from GitHub
* sh 'echo Hello' - Runs Linux commands inside agent. Equivalent to terminal command.
* echo "Build started" - Prints message in Jenkins console log.

3. The Challenging Part: withCredentials
   You need to login to Docker Hub, but you should NOT write password directly in code. Jenkins stores it securely.

Step-by-Step Meaning:
1. Store Secret in Jenkins with ID: dockerhub-token
2. withCredentials Block temporarily injects secret into environment variable.
string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')
* string = Type of secret (text token)
* credentialsId = Name used in Jenkins
* variable = Temporary variable name
3. What Happens Internally: Jenkins does DOCKER_TOKEN=your_actual_secret_value
4. Using It in sh: sh 'echo $DOCKER_TOKEN | docker login -u username --password-stdin'

Simplified Analogy:
* Locker (Jenkins Credentials) -> You ask by ID (dockerhub-token) -> Jenkins gives temporary key (DOCKER_TOKEN) -> You use it then it disappears.

4. Key Takeaways
   * pipeline -> stages -> stage -> steps = structure
   * sh = run commands
   * git = fetch code
   * withCredentials = securely use secrets
   * Secrets are temporary and protected